Artificial Intelligence In Medicine

Research paper

AI- Augmented DTN for resilient rural telepathology

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ABSTRACT

Rapid and accurate diagnosis of histopathological samples is critical for early detection of diseases such as cancer. However, rural healthcare settings often face challenges due to limited network infrastructure and lack of specialized pathologists. Telepathology offers a solution, but large high-resolution images are difficult to transmit over low-bandwidth networks. This work proposes an AI-augmented delay-tolerant telepathology framework to improve transmission efficiency and reliability. A deep learning classifier prioritizes biopsy images into high-priority and routine cases, enabling faster transmission of critical samples. A neural compression model reduces image size while preserving diagnostic features, achieving approximately 34 dB PSNR, 0.95 SSIM, and 0.0004 MSE. Delay-tolerant networking ensures reliable data delivery under unstable connectivity, achieving 100% transmission success with an average throughput of 264 KB/s and 88% data reduction. The classification model achieves 99% accuracy with an F1-score of 0.99 and AUC-ROC of 0.9995. The results demonstrate improved feasibility of telepathology in bandwidth-constrained environments.

1. Introduction

In current times, the practice of histopathology is crucial for the proper diagnosis of illnesses, including cancers, inflammation, and infections. The role of the histopathologist is to examine the tissues through a microscope in order to detect any abnormalities in the structures or molecular changes that are indicative of the disease process. Until now, the process of diagnosis has been done by physical inspection using a microscope in a laboratory setting. In most cases, the presence of skilled pathologists and laboratories is not equal in different geographic locations; hence, people living in rural areas have difficulties obtaining a diagnosis since their healthcare centres may lack the necessary skills or equipment to diagnose diseases in time.

The technology known as telepathology has proven to be a useful way of resolving this issue through digitization of pathology slides, allowing for their further remote analysis. Telepathology involves scanning of biopsy samples into high-resolution digital images and transferring them to the remote expert pathologists at specialized central hospitals. This approach facilitates collaboration between specialists who can provide assistance remotely to medical facilities that do not have sufficient local experience. As digital pathology becomes increasingly prevalent in today's medical practices, telepathology becomes an integral part of the telemedicine infrastructure.

At the same time, the introduction of telepathology to the real-world practice poses a range of challenges related to the use of these systems. Among the main obstacles to efficient operation is a considerable size of whole-slide images (WSIs). Modern digital slide scanners generate gigapixel WSI images which can exceed hundred megabytes per slide. The transfer of images of such size requires robust, high-speed internet connection which is often absent in many remote clinics in the countryside regions. Rural healthcare facilities often use mobile networks which can be unreliable and slow.

The recent developments in AI have also enabled many applications in improving medical imaging and diagnostic procedures. The use of deep learning algorithms has proven effective in various applications of medical imaging such as tumor localization, classification of tissues, and pathology images segmentation. Many studies have applied deep learning approaches to whole-slide image classification and computational pathology, revealing that computer-based diagnosis of certain tasks is similar to those performed by human professionals [2], [3], [5]. Moreover, deep learning frameworks have also been utilized in identifying cancer biomarkers and diagnosing breast cancer using histopathology images [8].

Federated learning is another major development in the realm of medical AI technology. The problem with traditional machine learning pipelines is that all the data from various hospitals needs to be centralized and processed on the same computer, which poses substantial risks to patient privacy. With federated learning, however, various institutions can train models locally while only sending model updates to the central server. Federated learning allows for developing collective models while respecting patient privacy at the same time. Recently, the effectiveness of federated learning techniques in solving computational pathology problems related to processing whole-slide images has been proven by numerous researchers [1]. Moreover, federated learning-based architectures for improving medical imaging segmentation have been discussed in several recent papers [4], [6], [7]. Finally, surveys of various federated learning systems have been conducted and published [12], [13], [26], [27].

While federated learning can be used to resolve privacy issues and facilitate distributed training, it is not applicable to resolving communication issues related to sending large medical images through unstable network connections. Healthcare facilities in rural areas can face problems with unstable network connectivity, long network latencies, and lack of network bandwidth. All these factors can result in failure of data transmission, inefficient usage of network bandwidth, and data delay. Conventional communication approaches relying on continuous network connection may not be efficient in such an environment.

Delay-Tolerant Networking (DTN) is a new type of communication model proposed for networks with intermittent connectivity. DTN communications are based on a store-carry-forward approach that allows storing data in intermediary nodes until the network becomes accessible. DTN is useful in situations when there is no constant connectivity between the two communicating entities. It can be successfully applied in remote locations, during disaster response, and in mobile sensors networks. Several recent publications prove the feasibility of delay-tolerant communication models in healthcare environments by providing reliable data delivery for healthcare purposes and facilitating remote medical services [9], [10], [11].

Aside from issues related to communication, another critical factor that hinders the implementation of telepathology systems is the enormous size of digital pathology images produced through whole slide scanning. As a result, there is a need for effective data compression techniques to minimize the data transfer between the network while retaining crucial elements for diagnosis within the images. Although conventional compression techniques like JPEG and JPEG2000 are commonly employed for medical images, these techniques often encounter limitations when dealing with intricate histopathology images. There have been studies on the use of deep learning algorithms to compress and reconstruct medical images by recognizing important features while discarding redundant parts of the image [16], [17].

It is important to ensure that the quality of the images produced during image compression remains high for the application in clinical settings. Some of the common methods of assessing the quality of image reconstruction include peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and mean squared error (MSE). PSNR is used to measure the signal strength compared to the noise level during the reconstruction process. The higher PSNR and SSIM levels together with lower MSE mean that the image quality of the compressed images is high enough for use in clinical settings.

Another trend in health care technologies is the adoption of edge computing and intelligent processing at the source of data generation. Edge health care systems provide medical devices and local processing units with the capability of performing preliminary analysis on the collected data before sending it to a central server. This strategy helps mitigate the traffic burden on the network and optimize the time of response by sending less raw data over the network. Edge cognitive computing paradigms have been suggested to facilitate real-time processing of health care data [25]. Likewise, edge-cloud cooperation strategies have been investigated to analyze health care data securely in distributed Internet of Things networks [24].

In addition, the increasing complexity of multimodal medical data has led researchers to explore the potential of representation learning algorithms designed to differentiate between modality-specific components and common anatomical components. Disentanglement learning has been used to distinguish between domain-specific differences and medically important information within medical imaging data, allowing machine learning algorithms to generalize across multiple datasets and various imaging settings [18]. In addition, cross-modality representation learning has been suggested for extracting common feature representations from various imaging modalities, enabling the transfer of knowledge across diverse medical data sources [19], [20]. These approaches can enhance the performance of telepathology solutions in practical clinical scenarios, where imaging data is likely to be heterogeneous due to different scanning equipment, imaging protocols, or patient demographics.

However, although significant advances have been achieved in recent years in terms of AI-enabled medical imaging, data compression technologies, and decentralized communication networks, most current investigations have explored these issues separately. An effective telepathology solution should incorporate all three aspects – advanced medical image analysis, data compression techniques, and communication networks – to enable seamless and reliable operation even under challenging circumstances.

Given these problems, the current research attempts to evaluate an AI-enhanced telepathology framework that combines automatic analysis of pathology images, high-performance image compression algorithms, and delay-tolerant communication protocols. The purpose of the new telepathology framework is to allow for the effective transfer and analysis of pathology images under conditions of limited availability of network infrastructure while maintaining the accuracy of diagnosis and good image quality. Utilizing modern advancements in the field of artificial intelligence, pathology image compression, and delay-tolerant networking, the framework can provide improved telepathology access in disadvantaged regions.

1.1 Research Gaps

However, despite advancements made in the realm of artificial intelligence and digital pathology techniques, some research gaps have been identified in relation to the creation of telepathology systems. For instance, there is a clear lack of effort aimed at tackling issues that emerge when trying to transmit huge pathological images through unreliable networks. Whole-slide images are massive in size, meaning that a mechanism needs to be created for their efficient compression and subsequent transfer.

Another important issue that should be considered involves the fact that many telemedicine systems make use of conventional networking protocols that presuppose stable connectivity. Such approaches are less applicable in the context of rural healthcare facilities due to poor or unstable networks available within such regions. Delay-tolerant networking has already been applied in various contexts but not in telepathology systems yet.

Thirdly, recent work on telepathology tends to study image processing, image compression, and communications separately instead of considering all three as an integral part of a system design that should consider the aspects of diagnostic correctness, efficiency of information transfer, and reliability of network communication at once.

Lastly, with a growing amount of medical data available from various institutions, issues related to machine learning on insufficient data become more apparent. Federated learning and representation disentanglement are techniques currently proposed for dealing with the problem, yet their implementation into telepathology networks is still a subject for future research.

1.2 Motivation

The rationale of this research is based on the need to create a robust framework for telepathology which will be capable of functioning in a bandwidth-limited environment. The use of modern technologies in artificial intelligence and image compression can help develop a telepathology system which would function well even under the conditions of low network connectivity.

These systems could be very helpful in expanding the availability of healthcare services since it is possible to establish connections between rural clinics and specialist hospitals. Timely and efficient diagnostic processes would enable faster diagnoses of diseases, leading to better outcomes and treatment plans. Additionally, the incorporation of intelligent data processing and networking techniques would help minimize network resources consumed by a telepathology system while maintaining the quality of transmitted medical images.

Overall, there is a great need to design integrated telepathology systems with AI technology for bridging the gap between urban medical facilities and rural areas.

1.3 Aim and Objectives

The key objective of this research is to create a robust and efficient telepathology system that can enable reliable diagnosis even under limited bandwidth connections. In many rural areas where infrastructure development is still a challenge, there are fewer numbers of specialists for analyzing such pathological cases, leading to delay in the process of diagnosing. Furthermore, the creation of large pathological images through digital scanning technologies has added another layer of complexity to the process, making the transfer of these pathological images quite cumbersome due to poor network conditions. Thus, the key objective of this research is to come up with an integrated approach to pathology image analysis and compression to facilitate reliable transmission.

In order to accomplish this objective, the study will focus on combining AI methodologies with effective networking methods in an attempt to overcome the technical difficulties of telepathology implementation in remote regions. This proposed design is based on the principles of image processing, effective communication, and automated pathology image classification. Thanks to the advancements in the field of deep learning and medical image processing, the methodology aims at reducing the size of the pathology image while preserving important information about its structure. Furthermore, delay-tolerant networking solutions will be used to guarantee the effectiveness of transferring medical data in case of unreliable networks.

The specific objectives of this research are outlined as follows:

1. **To develop an automated histopathology image classification model** capable of distinguishing between malignant and benign biopsy samples with high diagnostic accuracy, thereby assisting pathologists in prioritizing critical cases.

2. **To develop an effective compression algorithm for medical images using deep learning** approaches such that the amount of pathology images is reduced substantially without affecting the diagnosing features of the structure.

3. **To assess the quality of reconstructed medical images using popular measures** in medical imaging including Peak Signal to Noise ratio (PSNR), Structural Similarity Index (SSIM), and Mean Squared Error (MSE) to confirm that the compressing process does not affect the diagnosing capability of the images.

4. **To establish a delay tolerant data transfer system** that ensures that large sized medical images can be transmitted effectively in scenarios with poor connectivity and latency conditions**.**

5. **To examine the effectiveness of the system by carrying out various experiments** involving evaluating the classification accuracy, compressing effectiveness, and networking performance including throughput, latency, and success rate of data transmission.

6. **To demonstrate the feasibility of integrated AI-driven telepathology systems** for improving remote diagnostic capabilities and supporting healthcare services in underserved and rural regions.

With the help of these goals, this research strives to make contributions to the development of telepathology applications that will be able to perform efficiently in practical settings, where the available resources are not abundant. The combination of intelligent image analysis, effective image compression, and reliable communication systems could have a positive impact on the performance of such applications.

2. Background and Related Works

Recent technological advancements in artificial intelligence and digital medical solutions have brought major changes to the sphere of computational pathology and remote medical care. Due to the growing popularity of digital pathology solutions, the histopathological samples can be scanned to form high-definition WSIs and then analyzed automatically and remotely. Although recent advances have led to significant improvements in terms of diagnostics and medical studies, there is still room for improvement as regards the transmission and analysis of large-scale medical images in healthcare. This section summarizes the current state of literature on four main topics: computational pathology and WSI analysis, federated learning in medical imagery, delay-tolerant networking for healthcare communications, and deep learning-based medical image compression.

2.1 Computational Pathology and Whole-Slide Image Analysis

Recent developments in computational pathology have seen tremendous interest from various researchers owing to its capabilities in automating the analysis of pathologic samples. Whole-slide imaging technology entails scanning the tissues at an exceptionally high resolution to generate gigapixel images which capture the microscopic details. With the help of the resulting digital slides, various machine learning algorithms can be used to analyze the tissue morphology as well as detect abnormalities and help clinicians make diagnoses.

There have been many attempts to use deep learning models on whole-slide images. For instance, Zheng et al. suggested the design of a graph-transformer model for the classification of whole-slide images to improve upon the performance of previous approaches when dealing with histopathology images [2]. Another related study involved the development of a recalibrated multi-instance deep learning model for the analysis of gastric pathology image data, which led to improvements in tumor detection within a large dataset of gastric pathology images [3]. Additionally, Zhou & Lu developed hierarchical multiple-instance learning techniques that allow deep learning models to handle whole-slide images efficiently [5].

Moreover, it should be noted that the deep learning technique has found application even in the field of clinical practice, where it is used both for diagnosing breast cancer and for recognizing biomarkers. As an illustration, Aboudessouki et al. used deep learning to develop a tool for detecting molecular biomarkers in whole-slide pathology images [8]. It can be assumed that the growing importance of artificial intelligence in computational pathology is proved by such studies.

2.2 Federated Learning in Medical Imaging

Despite the efficiency of deep learning algorithms in analyzing pathological images, the amount of necessary data for training the algorithms is enormous and can be obtained only from multiple organizations. Medical data exchange among healthcare providers involves several challenges in terms of data security and privacy. Federated learning is considered an efficient solution for resolving the above-mentioned problems via cooperative training of models without any exchange of patients' personal information.

Parties engaged in federated learning create individual models based on private datasets and transmit only neural network weights to the central server. The server then generates a global model by combining the data of local models without any data exchange. As demonstrated by Lu et al., federated learning is able to address problems associated with computational pathology in gigapixel whole-slide images as effectively as traditional centralized algorithms [1].

In recent years, there has been a significant number of researches concerning federated learning methods in medical imaging. As examples, we could mention such works as the one carried out by Jiang et al., who proposed Inside-Out Personalization Federated Learning framework (IOP-FL), increasing segmentation accuracy through using personalization techniques as well as global models [4]. Another example is Li et al., who developed a transformer-based federated learning framework for prediction of breast cancer biomarkers on the basis of whole-slide image analysis [6]. Moreover, Zhang et al. designed FedSODA architecture specifically for histopathology segmentation tasks [7].

However, apart from the above-mentioned works, there were several review papers dedicated to the research of federated learning application in medicine. For example, Rauniyar et al. developed a taxonomy of federated learning approaches utilized in medicine, solving issues concerning data heterogeneity, communication efficiency, and scalability problems [12]. Additionally, Antunes et al. proposed some design concepts for federated learning frameworks for analyzing the medical data in a joint manner while retaining their confidentiality [13]. Also, some other surveys pointed out the relevance of federated learning in sensitive datasets applications, such as healthcare area [26], [27].

2.3 Delay-Tolerant Networking for Healthcare Communication

Despite all the advances made in AI in medicine and distributed learning architectures, there is still a problem in transmitting large medical images through unreliable network environments. Most remote healthcare centers have limited access to network connectivity because of their low bandwidth, high latency, and limited availability. Conventional communication mechanisms work on the assumption that all the connected devices are online at all times.

Delay-Tolerant Networks (DTNs) can serve as reliable mechanisms of communication in such settings due to their ability to provide connectivity in the absence of end-to-end connections. In DTNs, the store-and-carry forward approach is utilized, where messages get saved at various nodes until the right time for their transfer comes.

Spanakis and Sakkalis explored the possible implementation of delay-tolerant proxy service for healthcare communication infrastructure to justify the reliability of DTN technology for data transfer within healthcare communication channels [9]. Additionally, Rahman et al. discussed the possible effects of delay-tolerant communication strategies on the process of data gathering in sensor networks in order to justify the efficiency of implementing adaptive routing and buffering strategies [10]. Furthermore, More and Kale provided an exhaustive literature review regarding the importance of DTN in communication and monitoring infrastructures [11].

All these research works suggest that the implementation of DTN strategies in telemedicine and telepathology may help improve the reliability of the communication infrastructures.

2.4 Deep Learning–Based Medical Image Compression

Another issue related to telepathology technology is the large size of images generated through digital pathology scanning machines. Usually, the size of images generated from digital pathology scanners is relatively huge, with hundreds of megabytes making it highly necessary to optimize compression techniques to save network communication expenses. However, traditional compression algorithms, including JPEG and JPEG2000, have not managed to effectively balance compression efficiency and clinical usefulness of the compressed images.

Developments in deep learning technology have led to smart compression algorithms that can efficiently preserve clinical characteristics of images when compressing the size of images. According to Gupta et al., compression algorithms were designed using deep learning technology that helped extract clinical parts of images [16]. Xie et al. suggested another smart algorithm for compressing images for medical purposes using deep learning technology with transfer learning approach [17].

In addition to the use of compression techniques, some scholars tried applying representation learning and disentanglement for improvement of the process of reconstructing an image. Thus, Xie et al. proposed using domain adaptation approach based on the disentanglement learning technique for separating modality-specific features from general anatomy in medical images [18]. In addition, Reaungamornrat et al. demonstrated the opportunity to utilize disentangled representations for creating multi-modal medical images, which allowed a neural network to distinguish between the general anatomical structures and the unique properties of a particular modality [19]. In turn, Li et al. studied cross-modality representation learning methods used by models for extraction of common features in various imaging modalities [20].

These are just some examples of how deeply involved complex deep learning is getting in the medical sphere.

2.5 Edge Intelligence and Smart Healthcare Systems

The latest advancements in smart healthcare technologies have focused on the need for edge computing to support real-time data processing for clinical purposes. Edge computing is a paradigm which enables carrying out computations close to the origin of the data, hence decreasing network latency and making processing faster. Using edge computing in health care organizations means performing basic analysis on medical images before data can be sent to centralized servers.

Gupta has presented a collaborative framework between edge and cloud computing for implementing IoT-enabled security in health care settings [24]. Likewise, Chen et al. developed a smart edge cognitive computing framework for healthcare applications [25]. This illustrates the possibilities and benefits that exist by combining edge computing with telemedicine systems.

the network infrastructure. All three modules combine to increase diagnostic efficiency while making sure that communication of pathology images is made reliably.

The first step involves acquiring digital images of pathology samples from biopsy specimens. After obtaining these images, a deep learning-based image classifier is used to classify whether the tissue sample is benign or malignant. Once classification is completed, an intelligent compression module uses deep learning-based methods to compress pathology images by removing any irrelevant features from the image. Compression ensures that images can be transferred using delay-tolerant networking techniques.

3.1 System Architecture Overview

Architecture of the proposed scheme is as follows:

1. Image Acquisition – High-quality imaging of histopathology slides through digital pathology scanners.
2. Image Classification – Pathology images are classified using deep learning techniques to differentiate between cancerous and non-cancerous tissues.
3. Image Compression – Deep Neural Network compression technique compresses images without compromising on clinically relevant features.
4. Delay-Tolerant Communication – The compressed images are divided into blocks and communicated through delay-tolerant communication protocols.
5. Image Reconstruction & Diagnosis – Image reconstruction and diagnosis is carried out at the receiver node.

The purpose of the pipeline is to facilitate efficient processing and transmission of pathology images among remote healthcare facilities and diagnosis centres

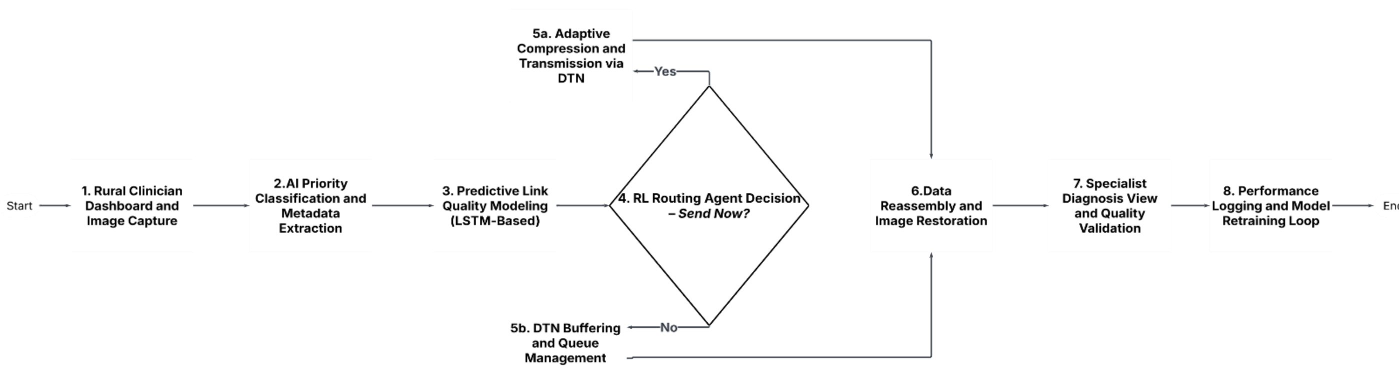

Fig. 1. We present the proposed AI-augmented telepathology workflow integrating priority-based classification, adaptive compression, delay-tolerant networking, and diagnostic analysis.

2.6 Summary of Related Works

From the review of literature above, it is clear that considerable progress has been made in the areas of computational pathology, federated learning, delay-tolerant networking, and deep learning-based medical image compression. Nevertheless, most current works deal with specific elements of telepathology systems without offering comprehensive systems that cater to all the requirements of telepathology. Most works in computational pathology concentrate on improving diagnostic capabilities, while those in networking are mainly concerned with improving network reliability. In similar fashion, works in image compression do not consider transmission needs when designing their schemes.

There is therefore a need for comprehensive telepathology framework that incorporates intelligent medical image analysis, effective compression algorithms, and robust transmission techniques in order to deliver reliable telepathology service in bandwidth-limited environments. Such a system is the objective of the proposed telepathology framework.

3. Methodology

The methodology section below discusses the telepathology system methodology, which is designed to facilitate reliable diagnosis and transfer of pathology images. The process consists of three main modules: automatic histopathology image classification, intelligent compression of the image through deep learning, and delay-tolerant transfer of pathology images over

3.2 Histopathology Image Classification

The automated classification of the pathology images is helpful in the identification of the high risk cases. CNNs are often employed in medical image classification due to their ability to learn hierarchical image features automatically.

I will use “I” to denote the input pathology image. Through the classification model, the input image is analysed to generate a class prediction.

Classification model representation

Word equation format

$$y = f(I, \theta) \tag{1}$$

Where:

I = input histopathology image

θ = model parameters

f = trained neural network model

y = predicted class label (malignant or benign)

Classification Evaluation Metrics:

Accuracy

Word equation format:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{2}$$

Where:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Precision

$$Precision = \frac{TP}{(TP + FP)} \tag{3}$$

Precision measures how many predicted positive samples are actually positive.

Recall (Sensitivity)

$$Recall = \frac{TP}{(TP + FN)} \tag{4}$$

Recall measures how effectively the model identifies positive cases.

F1 Score

$$F1\ Score = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)} \tag{5}$$

The F1 score provides a balanced evaluation of classification performance.

3.3 Deep Learning–Based Image Compression

Digital pathology images produced by the whole-slide scanner are very large. Hence, it is imperative to develop efficient compression methods to minimize the bandwidth for transmitting such images over a network, without compromising their diagnostic quality.

The suggested compression framework leverages an advanced deep learning model known as the Residual U-Net. The model comprises of two major parts:

- Encoder
- Decoder

The encoder takes the input image and compresses it into a latent space, whereas the decoder decompresses the latent representation to produce the output image.

Encoder Representation

$$z = E(I) \tag{6}$$

Where:

I = original pathology image

E = encoder function

z = compressed latent representation

Decoder Representation

$$\hat{I} = D(z) \tag{7}$$

Where:

D = decoder function

Î = reconstructed image

Compression Quality Metrics:

Mean Squared Error (MSE)

$$MSE = \left(\frac{1}{N}\right) \times \Sigma \left(Ii - \hat{I}i\right)^2 \tag{8}$$

Where:

Ii = pixel value of original image

Îi = pixel value of reconstructed image

N = total number of pixels

MSE measures the reconstruction error between original and reconstructed images.

Peak Signal-to-Noise Ratio (PSNR)

$$PSNR = 10 \times log10 \left(\frac{MAX^2}{MSE}\right) \tag{9}$$

Where:

MAX = maximum possible pixel value of the image

MSE = mean squared error

Higher PSNR values indicate better reconstruction quality.

Structural Similarity Index (SSIM)

$$SSIM(x, y) = \frac{\left((2\mu x\, \mu y + C1)(2\sigma xy + C2)\right)}{\left((\mu x^2 + \mu y^2 + C1)(\sigma x^2 + \sigma y^2 + C2)\right)} \tag{10}$$

Where:

x = original image

y = reconstructed image

μx , μy = mean intensity values

σx , σy = standard deviations

σxy = covariance

SSIM measures perceptual similarity between images.

3.4 Delay-Tolerant Data Transmission

Reliable delivery of medical images is problematic in networks with unreliable network connectivity. The solution for this issue has been provided by Delay-Tolerant Networking (DTN), where the method of storing and carrying data packets forward is adopted.

The data packets are stored in intermediate nodes until an adequate communication channel arises. Pathology images of large size are segmented into small chunks referred to as bundles.

Network Performance Metrics

Throughput

$$Throughput = \frac{Total\ Data\ Transferred}{Transmission\ Time} \tag{11}$$

Throughput measures the efficiency of data transfer across the network.

Transmission Success Rate

$$Success\ Rate = \frac{Successful\ Transmissions}{Total\ Transmission\ Attempts} \tag{12}$$

This metric evaluates the reliability of the data transmission process.

End-to-End Latency

$$Latency = Time_{received} - Time_{sent} \quad (13)$$

Latency represents the total delay required for data to travel from sender to receiver.

3.5 Integrated Telepathology Framework

This methodology combines the use of image classification, intelligent compression, and delay tolerant networking in a single telepathology application. Image classification helps in automatically identifying cases that are malignant, thus helping clinicians to identify cases urgently requiring diagnosis. Intelligent compression compresses the images, ensuring that their structure is preserved for diagnosis purposes. Delay tolerant networking will ensure that the images are transferred effectively regardless of network conditions.

With all the above features combined, this methodology presents an effective approach for remote pathology diagnosis especially in areas with limited resources.

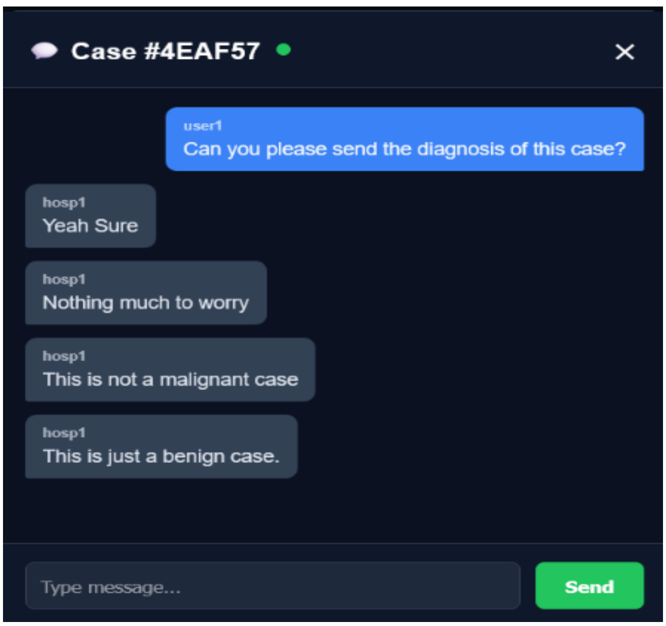

Fig. 2a. We present the diagnosis request interface at the rural healthcare centre, where clinicians initiate a request for diagnostic results after transmitting pathology images.

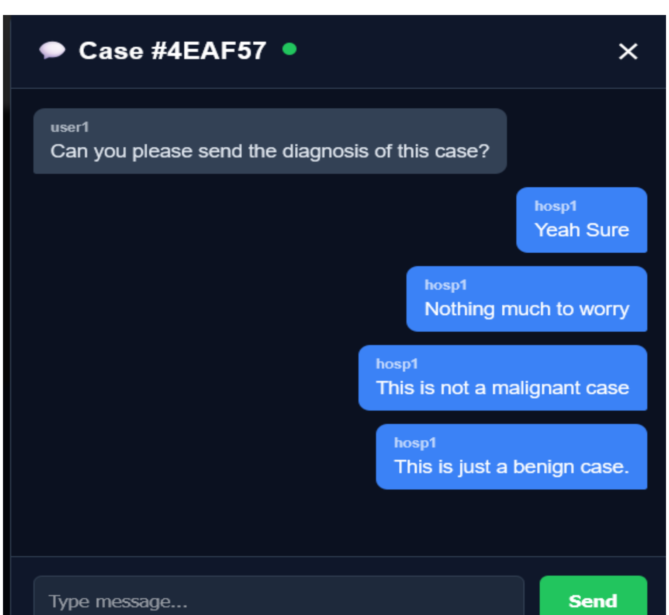

Fig. 2b. We present the hospital-side diagnostic response interface displaying AI-assisted classification results indicating a benign case.

4. Experimental Setup and Evaluation

In this part, we will give a detailed introduction of the experimental design scheme for testing the performance of the proposed AI-aided telepathology system. The experiments were conducted to examine the following three key areas of this system: diagnosis classification accuracy, image compression and decompression results, and communication channel transmission efficiency under bandwidth-limited conditions. Due to the combination of several different technologies in this proposed telepathology solution, including artificial intelligence, medical image compression, and delay tolerant networks, it is necessary to test each technology separately and simultaneously.

The experimental study thus considers the following three areas of performance assessment. Firstly, the classification ability of the algorithm will be analyzed by utilizing common metrics for measuring image classifications in medicine such as accuracy, precision, recall, F1 score, and AUC-ROC. The analysis will highlight the efficacy of the proposed automatic pathology analysis process in identifying benign and malignant tissue samples. Secondly, the compression algorithm will be analyzed through metrics used in the reconstruction of medical images such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). The metrics will verify that the proposed compression process does not distort key information that is necessary for medical diagnostics. Thirdly, the efficacy and efficiency of the proposed transmission architecture will be tested by applying networking metrics such as transmission success rate, network throughput, and latency.

Through the analysis of the above factors, the experimental evaluation intends to prove the suitability of the proposed telepathology framework for facilitating the effective analysis and transmission of medical images within healthcare institutions without robust network facilities.

4.1 Dataset and Preprocessing

The experimental evaluation was performed based on a database consisting of digitized images of histological preparations obtained from biopsies. Digital images of histological preparation carry information about the structure represented by cellular morphological patterns as well as tissue structure and pathology of interest. Therefore, automatic recognition of the images is a difficult task that requires detection of the local cellular pattern as well as tissue level pattern information. The database used in this experiment is the BreakHis Database [31] that consists of images of histological preparations of benign and malignant breast cancers.

The database used for this experiment consists of 1563 images of labeled pathologies including 477 samples of benign and 1086 samples of malignant pathologies.

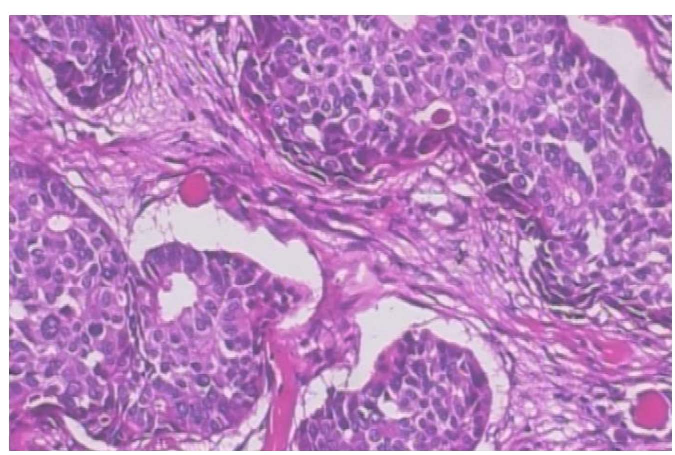

Fig. 3. We present a sample malignant histopathology image used in the dataset, showing abnormal cellular structures and dense tissue morphology associated with cancer.

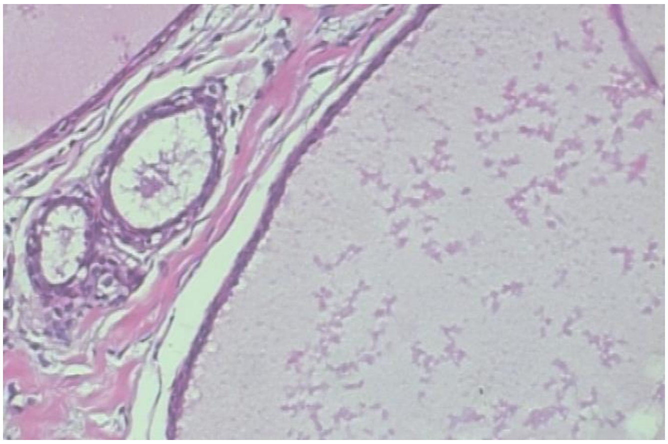

Fig. 3. We present a sample malignant histopathology image used in the dataset, showing abnormal cellular structures and dense tissue morphology associated with cancer.

Prior to training the two models, a number of image preprocessing steps were performed on the raw pathology images. These include the processes of normalizing, resizing, and de-noising images. Histopathology images tend to be acquired at very high resolutions; hence, the cost of processing the entire slide would prove expensive for model training. Therefore, the size of each image was resized before training.

Apart from the simple preprocessing step that was carried out, additional steps were done through the use of data augmentation. Data augmentation helps the deep learning models generalize better when faced with unseen pathology images by introducing more variance to the training samples. The types of data augmentations performed included random rotation, random flips, and random crops of images.

The process of normalization was also used in order to normalize the values of pixel intensities in terms of their range. The use of normalization accelerates the process of learning neural networks and also makes input data compatible with the model's initialization parameters.

By using the above-mentioned techniques, the prepared dataset became ready for the training of both classification and compression parts of the system.

4.2 Classification Model Evaluation

The classification model for histopathological diagnosis was tested by various well-known performance measures that are applied in the field of medical imaging analysis. The objective of the classification model is to be able to classify histopathology tissue images as either benign or malignant based on certain features present in the images.

Accuracy refers to the percentage of correct classification within the entire set of samples in the data set. Accuracy is a good measure of the performance of the model; however, it does not necessarily indicate the performance of the system itself.

Thus, supplementary measures like precision, recall, and F1-score were utilized for the evaluation of the classification performance. Precision refers to the ratio of malignant samples among all the predicted malignant samples. This measure represents the accuracy of positive cases. Recall, which is also known as sensitivity, is another measure that reflects the efficiency of the model in recognizing the malignant samples. High values of recall are especially important in clinical practice since overlooking a malignant sample can lead to serious complications.

F1-score is an integrated measure based on precision and recall measures. It calculates the harmonic mean of the mentioned values. Such a measure allows achieving a more balanced evaluation of the classifier's performance.

Finally, to test the ability of the classification model to discriminate between malignant and benign cases, a supplementary measure called ROC curve was utilized. The ROC curve visualizes the trade-off between the rates of true positives and false positives depending on the threshold of classification. Additionally, AUC-ROC was calculated as a summary measure representing the quality of discrimination of the classifier. A higher value of AUC-ROC corresponds to better performance of the classifier.

The findings from the experiment show that the proposed classification algorithm can diagnose malignant pathology cases with an accuracy of about 99%. The precision and recall scores of the classification algorithm are nearly 0.99 for both classes. The F1-score and AUC-ROC scores are 0.99 and 0.9995, respectively.

This shows that the classification algorithm is highly accurate in diagnosing malignant pathology cases.

4.3 Image Compression Evaluation

Transmission of pathology images in an efficient manner is one of the crucial needs for telepathology applications that run under low-bandwidth networking. The whole-slide pathology images are often characterized by very high resolution as well as huge file sizes, sometimes running up to hundreds of megabytes per slide. Efficient compression mechanisms would make their transmission difficult otherwise.

In order to solve this problem, the presented research proposes the implementation of a deep learning-based compression algorithm that shrinks the size of images while retaining diagnostic features.

The performance of the compression algorithm was assessed in terms of three major reconstruction metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM).

Mean Squared Error quantifies the average square error between the original image and the reconstructed image obtained from the compression algorithm. Lower values of this metric mean that the reconstruction error is lower and thus the efficiency of the compression process is higher.

Peak Signal-to-Noise Ratio defines the ratio between the power of the original image signal and the noise added to the compressed image. High PSNR indicates a high-quality reconstruction. Acceptable values for medical imaging applications are greater than 30 dB.

Structural Similarity Index evaluates the perceptual similarities between two images by measuring their structural components, including brightness, contrast, and structure. This metric ranges from 0 to 1, where a value of 1 means that two images have strong structural similarities.

The outcomes of experiments prove that the suggested compression model is able to provide an average value for the PSNR index equal to about 34 dB, SSIM equal to about 0.955, and a minimal MSE value of about 0.0004. This proves that the compression model is effective enough in preserving the key structural elements in pathology images.

Additionally, the high SSIM value shows that the images reconstructed after compression are perceptually similar to the original ones, which guarantees the clear visibility of diagnostic features like cells.

4.4 Network Transmission Evaluation

Besides classification accuracy and compression effectiveness, the reliability of data transfer in telepathology technology was assessed based on networking parameters related to delay tolerant data transmission. The reliable transfer of data for medical imaging is especially crucial in areas where the networking conditions are unstable.

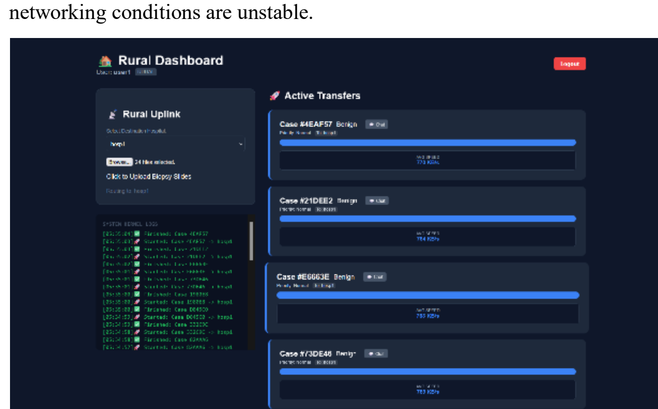

Fig. 5a. We present the rural dashboard interface for uploading biopsy images and monitoring active data transmissions.

Performance of the suggested transmission method was assessed by considering three main factors, namely, transmission success rate, network throughput, and latency.

Transmission success rate is the ratio of successful deliveries of image packets over total transmissions attempted by the system. High transmission success rate implies reliable delivery of data despite limited network availability.

Network throughput is the measurement of the rate at which data are successfully delivered through the network. Higher throughput figures imply that the available bandwidth has been used efficiently.

Latency refers to the total amount of time taken for an image packet to be sent from one point of origin to the destination. Lower latency rates imply improved transmission times.

According to the results obtained from experimentally evaluating the system, it has a transmission **success rate of 100%,** implying that all transmissions have succeeded. Network throughput was measured to be about **264.67 KB/s**, while latency was calculated to be about **1.07 seconds**.

Fig. 5b. We present the hospital command center dashboard displaying incoming cases organized in a priority queue along with system logs.

These findings show that the delay-tolerant communication scheme successfully facilitates dependable transmission of compressed pathology images in conditions characterized by low network connectivity. The following diagrams present the real-time monitoring dashboards for the suggested telepathology framework. The dashboard on the left represents the rural interface where biopsy images can be uploaded and monitored during transmission. On the other hand, the dashboard on the right represents the hospital interface that presents incoming cases in order of priority as well as real-time system logs and transfer rates. The rural dashboard allows the doctors to upload the images and track their transfer process as presented in Fig. 5a. It also contains information concerning the logs of the system as well as the progress of the transfer process. Similarly, the hospital dashboard in Fig. 5b presents the incoming cases and also offers the logs of the system in real time.

4.5 Scalability and Stress Testing

Additional tests were undertaken in order to check the efficiency and scalability of the proposed telepathology system through performing stress testing for transmission scenarios with several simultaneous image transfers.

In particular, the stress testing was conducted for transmission scenarios with 5, 10, and 20 simultaneous image transfers. The following parameters were measured during each experiment: transmission success ratio, average decision latency, maximum latency, and overall time of image transmission.

From the analysis of results presented above, it is possible to make conclusions concerning the successful operation of the proposed telepathology system. Despite additional loads on the network associated with simultaneous image transfers, all the data was transferred successfully. In addition, even at increased latency, it is still possible to transmit several images simultaneously and establish reliable communication.

Overall, the use of intelligent compression methods and delay-tolerant networking approach provides the ability to achieve efficient scaling of the telepathology system.

On the whole, the experimental testing shows that the suggested framework performs well in terms of diagnostic classification, image compression, and transmission. It proves that there is a possibility to implement AI-enhanced telepathology systems even in remote areas where networks can be weak.

5. Results and Discussion

This section discusses the results achieved after conducting the experiment on the performance of the proposed AI-assisted telepathology framework. The purpose of this evaluation is to assess the performance of the framework in terms of three aspects, which include the accuracy of diagnostic classifications, the quality of medical image compression, and effectiveness of data transmission in areas where the network connectivity is constrained. Given the nature of the proposed framework, which incorporates different technologies such as deep learning based classification, neural compression, and delay tolerant networking, it is important to assess not only the performance of individual components but also that of the overall system.

As seen in the results, the proposed telepathology framework performs well in classifying malignant pathology cases, compression without loss of structural information, and reliable transmission of images regardless of the limitations in terms of network connectivity. This section offers a more detailed explanation of the results achieved through the experiment on the proposed framework's classification capabilities, compression efficiency, and network transmission performance.

5.1 Classification Performance Analysis

The classification module within the suggested framework is responsible for recognizing whether a histopathology image contains benign or malignant tissues. An automated classification system is crucial for telepathology frameworks since it enables an early detection of any critical cases as well as prioritizes work for the pathologist. The application of deep learning–based classification systems is highly effective for medical imaging analysis owing to hierarchical features extraction.

In this study, the proposed classification model was trained and evaluated with the aid of the dataset that was created using the histopathology images. The dataset comprised 1563 samples where all the images were either benign or malignant tissues. Evaluation of the learning capacity and behavior of the model was conducted by evaluating it at different training epochs.

The training results show that the proposed model converges rapidly and continuously improves during the training process. The initial training results showed that the baseline CNN model performed with an accuracy of about 77.4%, while the accuracy of the proposed model stood at about 89.1%. The difference in the early accuracy shows that the proposed model learns better than the baseline model.

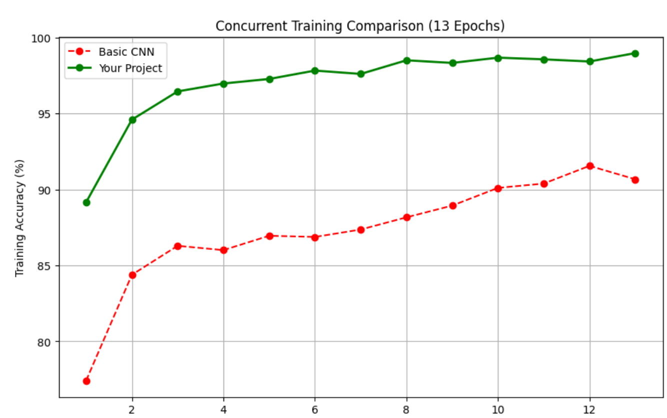

Fig. 6. We present the training accuracy comparison between the baseline CNN and the proposed classification model over multiple epochs.

In the course of the training, both networks demonstrated improvements in classification efficiency; however, the proposed network outperformed the baseline CNN model throughout all epochs. In particular, during the last training epoch, the baseline CNN model had an accuracy of 90.67%, while the proposed model showed an accuracy of approximately 98.98% for the same epoch. This means that there was more than 8% increase in classification accuracy when the proposed algorithm was used. As seen in Fig. 6, the proposed classification network constantly outperforms the baseline CNN throughout all training epochs. Moreover, the model demonstrates quicker convergence and greater accuracy in training.

The superior results of the classification network can be explained by the possibility of detecting complicated spatial relations in histopathological images. The structure of the tissue samples differs based on whether the patient has malignant or benign cancer; therefore, deep learning models trained to identify multi-scale features are effective for diagnostics.

In order to measure the performance of the classification algorithm in more detail, a classification report that includes precision, recall, and F1 scores is provided as standard evaluation measures.

The results revealed that the classification model exhibits an approximately 0.97 precision score for benign data and a 1.00 precision score for malignant data. Having high precision means that very low numbers of false positive predictions will occur; hence, any prediction about malignancy will be highly reliable. It should be mentioned that such an aspect is especially important in the clinical setting, as false alarms might require additional testing procedures.

High recall values (approximately 0.99) for each class have been observed, which implies that the classification model detects all types of cancer samples with very low error rates. Having high recall is especially important for cancer-related tasks due to the possibility of serious medical consequences if a malignant tumor is overlooked.

Finally, the F1 score, which reflects the balance between precision and recall, was calculated as 0.98 for benign samples and 0.99 for malignant samples.

An additional useful statistic utilized to assess classification accuracy is the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An ROC curve shows how sensitive a classifier is to various classification thresholds and depicts the correlation between true positive and false positive rates. The value of AUC serves as a measure that reflects the ability of a classifier to discriminate between classes.

In the present case, the proposed classification model has obtained a score of AUC-ROC equal to roughly 0.9995, indicating near-perfect discrimination between malignant tumors and benign masses. High values of AUC show that there is an excellent discriminative power of the model in question.

Thus, overall, classification results prove that a proposed algorithm allows for proper analysis of histopathology images and accurate detection of malignant tumor samples. This aspect is highly important for telepathology applications since automatic prioritization of such patients is crucial for efficient clinical practice.

5.2 Image Compression and Reconstruction Results

Highly efficient compression of pathology images is one of the requirements for a successful telepathology solution working in a constrained network bandwidth environment. Large whole-slide images produced by digital pathology scanners can easily have several hundreds of megabytes each. Transmitting such large images through low-bandwidth networks without proper compression techniques would be very ineffective.

The compression model designed within the proposed framework leverages a deep neural network trained to represent pathology images efficiently but still retain key structural features needed for medical diagnosis purposes. The performance of the proposed compression model has been estimated with image reconstruction measures such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). Fig. 7a shows how the model has steadily improved its reconstruction capabilities over training iterations. It is clear from this plot that PSNR and SSIM measures keep rising, whereas MSE is steadily decreasing.

In particular, throughout the training process, the proposed model kept improving its reconstruction performance in terms of the metrics mentioned above. As the training progressed, the reconstruction error became consistently smaller. Compared to a simple convolutional compression model, our architecture achieved significantly better results in terms of reconstruction error.

```
Epoch 1/20 | ResUNet -> PSNR: 32.10dB, SSIM: 0.9419, MSE: 0.0006 | Time: 5.6m
Epoch 2/20 | ResUNet -> PSNR: 32.52dB, SSIM: 0.9465, MSE: 0.0006 | Time: 5.6m
Epoch 3/20 | ResUNet -> PSNR: 33.18dB, SSIM: 0.9494, MSE: 0.0005 | Time: 5.6m
Epoch 4/20 | ResUNet -> PSNR: 32.71dB, SSIM: 0.9492, MSE: 0.0005 | Time: 5.6m
Epoch 5/20 | ResUNet -> PSNR: 33.63dB, SSIM: 0.9525, MSE: 0.0004 | Time: 5.6m
Epoch 6/20 | ResUNet -> PSNR: 33.46dB, SSIM: 0.9511, MSE: 0.0005 | Time: 5.6m
Epoch 7/20 | ResUNet -> PSNR: 32.59dB, SSIM: 0.9521, MSE: 0.0006 | Time: 5.6m
Epoch 8/20 | ResUNet -> PSNR: 33.51dB, SSIM: 0.9516, MSE: 0.0004 | Time: 5.6m
Epoch 9/20 | ResUNet -> PSNR: 33.30dB, SSIM: 0.9532, MSE: 0.0005 | Time: 5.6m
Epoch 10/20 | ResUNet -> PSNR: 33.81dB, SSIM: 0.9527, MSE: 0.0004 | Time: 5.6m
Epoch 11/20 | ResUNet -> PSNR: 34.11dB, SSIM: 0.9536, MSE: 0.0004 | Time: 5.6m
Epoch 12/20 | ResUNet -> PSNR: 33.93dB, SSIM: 0.9536, MSE: 0.0004 | Time: 5.6m
Epoch 13/20 | ResUNet -> PSNR: 33.68dB, SSIM: 0.9528, MSE: 0.0004 | Time: 5.6m
Epoch 14/20 | ResUNet -> PSNR: 33.91dB, SSIM: 0.9543, MSE: 0.0004 | Time: 5.6m
Epoch 15/20 | ResUNet -> PSNR: 34.24dB, SSIM: 0.9544, MSE: 0.0004 | Time: 5.6m
Epoch 16/20 | ResUNet -> PSNR: 34.03dB, SSIM: 0.9549, MSE: 0.0004 | Time: 5.6m
Epoch 17/20 | ResUNet -> PSNR: 33.66dB, SSIM: 0.9542, MSE: 0.0004 | Time: 5.6m
Epoch 18/20 | ResUNet -> PSNR: 34.14dB, SSIM: 0.9549, MSE: 0.0004 | Time: 5.6m
Epoch 19/20 | ResUNet -> PSNR: 33.81dB, SSIM: 0.9539, MSE: 0.0004 | Time: 5.6m
Epoch 20/20 | ResUNet -> PSNR: 34.31dB, SSIM: 0.9555, MSE: 0.0004 | Time: 5.6m
```

Fig. 7a. We present the epoch-wise performance of the compression model showing PSNR, SSIM, and MSE values during training.

The results of the last reconstruction have shown that the compression method produced the mean value of MSE that was about 0.0004. Such low error rate implies that images after their reconstruction were quite similar to the initial ones in terms of pixels.

The value of PSNR generated by the compression model was equal to about 34.31 dB. For medical imaging, the accepted limit of this indicator was 30 dB, which meant that it was safe to consider image quality good when PSNR was more than 30 dB. Consequently, it could be stated that the value produced was sufficient to ensure image fidelity after compression.

One more parameter, which was used to evaluate the quality of the reconstructed images, was the Structural Similarity Index (SSIM). While PSNR measured errors of pixel intensities between the original and the processed image, SSIM assessed structural similarities based on luminance, contrast, and texture patterns.

In our case, SSIM had been equal to approximately 0.9555. It could be assumed that images had retained most of the structural information after being compressed, and all important features, like cell structure and tissue pattern, had been kept intact.

Comparison of the performance of the compression model with the proposed framework reveals that there is a significant improvement in PSNR and SSIM. In this case, while the existing compression model has managed to obtain PSNR of around 25 dB and SSIM less than 0.90, the proposed model performed better.

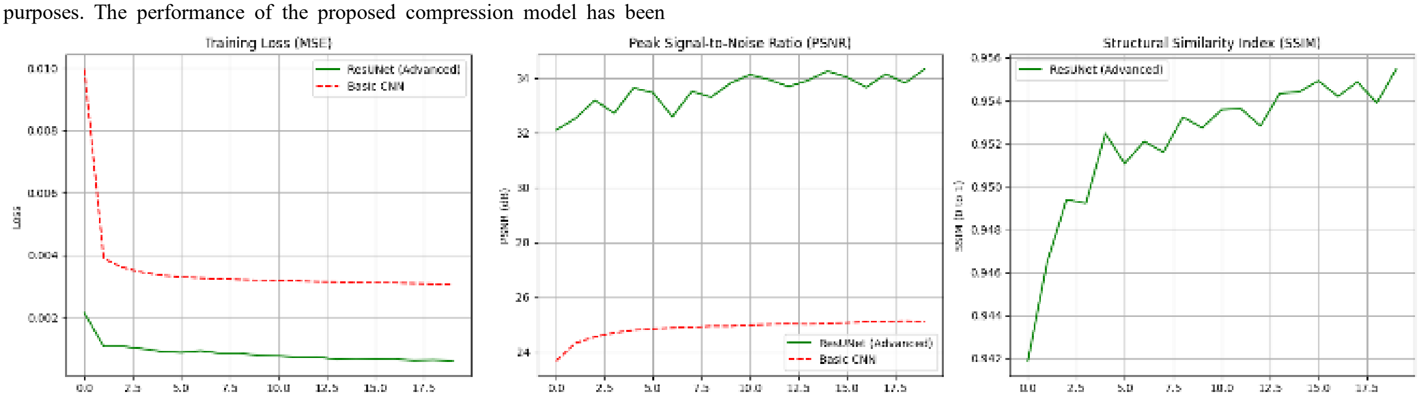

Fig. 7b. We present the compression model performance, showing training loss, PSNR, and SSIM trends over multiple epochs.

The above observations confirm that the compression model has been effective in reducing the size of the images without compromising the diagnostic value of visual information contained in images.

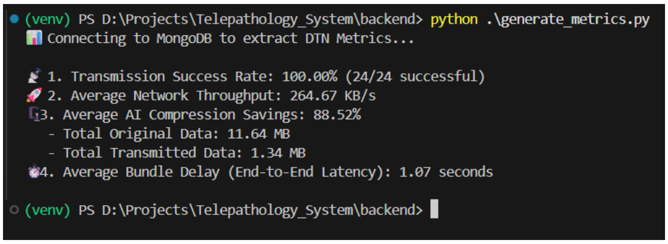

Fig. 8. We present the network transmission performance metrics, including success rate, throughput, compression savings, and end-to-end latency.

5.3 Network Transmission Performance

Reliability of the communication system is another aspect that should be considered in telepathology along with the performance in terms of classification and compression. Network bandwidth, connectivity problems, and communication availability are common issues in a rural healthcare environment. In such conditions, conventional data transmission procedures can become problematic because of unreliable transmissions.

To overcome the described problem, the suggested approach is based on delay tolerant networking methods that make it possible to transmit data reliably through an unstable network environment. Communication system performance was analyzed based on several networking parameters such as transmission success rate, network throughput, and latency.

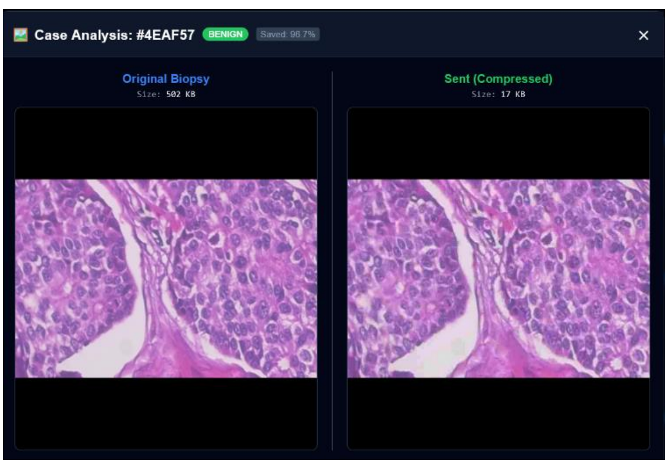

Fig. 9a. We present the compression stage showing the original biopsy image and its compressed version with reduced data size for transmission.

According to experimental findings, the suggested communication system provides 100% transmission success rate meaning that all transmitted image bundles were received by the receiver. Thus, the reliability of the system can be claimed as confirmed by the experimental research.

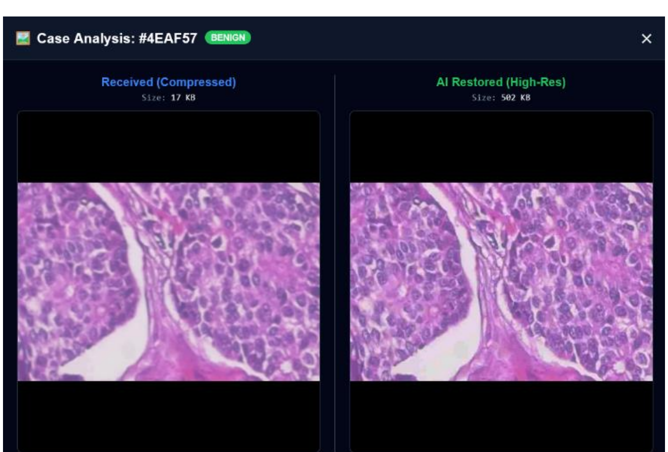

Fig. 9b. We present the reconstruction stage where the received compressed image is restored using the AI-based model to recover high-resolution diagnostic details.

The average throughput achieved in the process of testing was approximately equal to 264.67 KB/s. Although this figure seems quite low for high-speed communication, it corresponds to real-life rural conditions that can be faced while using the telepathology tool. The use of proposed compression model allows for reducing the amount of transmitted data considerably.

The average measured delay during our experiment was around 1.07 seconds. Latency involves the time taken to compress the image, transmit the data, and reconstruct the pathology image. The system still retains a low latency despite the extra time needed for the processes of compressing and decompressing the pathology image.

5.4 Scalability Analysis

In order to determine the scalability of the proposed framework, additional stress tests were performed using multiple image transfers at once. Such tests were performed simulating cases in which pathology images are sent simultaneously from several clinical locations. The following stress tests involved simultaneous transmission of five, ten, and twenty image files. During each test case, the transmission rate, latency, and time were measured.

It is seen that the system performs quite well in terms of scalability even when the amount of network traffic increases. All of the transmission cases are finished without errors and thus proving the effectiveness of the delay-tolerant protocol used. While the latency value increases with an increase in the number of simultaneous transmissions, the system can still cope with such traffic effectively.

Thus, it may be stated that the presented framework can be successfully scaled in the context of distributed telepathology systems.

5.5 Overall System Evaluation

The findings of the experiments on classification, compression, and networking confirm the efficient performance of the proposed telepathology approach in relation to all key evaluation metrics. Namely, the classification model generates highly accurate diagnostic predictions, the compression algorithm enables effective downsizing of image dimensions while maintaining their structural integrity, and the delay-tolerant networking approach guarantees successful data transmission in terms of limited network bandwidth.

In turn, the integration of these technologies into a single system makes it possible to develop a viable solution for telepathology diagnostics in underserved and remote healthcare areas. Automatic classification, efficient compression, and reliable data transmission through unstable communication channels are vital steps towards expanding the capabilities of telepathology technology and making its benefits available to patients from rural locations.

Consequently, the application of artificial intelligence-based approaches to telepathology has vast potential for facilitating diagnostics in remote areas.

6. Ablation Study

Ablation study was performed in order to quantify the effect of individual component in the framework on the performance of the overall telepathology framework. In the field of machine learning and system design, ablation studies are performed to analyze the effects of individual components in the model. It is possible to quantitatively evaluate the effect of various components of the model through the process of removing individual components. The effects can be measured in terms of image classification, reconstruction accuracy, and network communication efficiency.

The proposed telepathology framework consists of three main components. First, a deep learning-based image classification model was used for analyzing images from histopathological examinations. Next, a neural image compression model was utilized for compressing pathology images into smaller sizes without compromising image quality. Finally, the use of a delay-tolerant networking technique was proposed for efficient transfer of images.

6.1 Classification Model Comparison

In the first section of the ablation study, the classification model utilized in the analysis of the histopathology images will be considered. The proposed framework was tested against the baseline convolutional neural network (CNN). It aimed to demonstrate the superiority of the framework in terms of diagnosis accuracy and learning efficiency. Both the proposed framework and the CNN were trained using the same dataset, which consisted of benign and malignant histopathology images.

The results indicate that the proposed classification model exhibits superior performance compared to the baseline CNN in the process of training. At the initial training stages, the baseline CNN demonstrated a classification accuracy of 77.4%, while the proposed classification model exhibited an accuracy of 89.1%. Such results imply that the proposed classification model learns the pathological features more effectively at the early stages of training.

Both models improved their performances as the training process continued; nonetheless, the proposed model remained better in terms of accuracy. As the training ended, the accuracy score of the baseline CNN model was estimated to be 90.67%, while the accuracy of the proposed model was estimated to be 98.98%. The difference in accuracy of over 8% indicates the efficiency of the new feature extraction algorithm.

The classification report results further support the fact that the proposed model outperforms its counterpart. The precision scores of the proposed model for benign and malignant classes were approximately 0.97 and 1.00, respectively. These scores imply that the system generates few erroneous positive predictions.

Moreover, the recall scores for the same classes were approximately 0.99 each. The proposed model's precision and recall scores indicate its capacity to recognize almost all samples belonging to the malignant class. Finally, the F1-scores for benign and malignant classes were around 0.98 and 0.99, respectively.

Additionally, the proposed model generated an AUC-ROC score of 0.9995, which implies almost perfect separation between two diagnostic classes.

The above findings indicate that the better classification network architecture is a vital component in improving the diagnosis of the telepathology system designed. The differences in performance between the basic and proposed classification networks are illustrated in Table 1.

Table 1. Comparison of classification performance between baseline CNN and proposed model.

Model	Accuracy (%)	Precision	Recall	F1 Score	AUC-ROC
Baseline CNN	90.67	0.91	0.90	0.90	0.95
Proposed Model	**98.98**	**0.99**	**0.99**	**0.99**	**0.9995**

6.2 Compression Model Evaluation

The second section of the ablation study aims at measuring the efficacy of the proposed compression algorithm based on deep learning techniques.

It is crucial to have an efficient compression algorithm for the pathology images for telepathology purposes as the whole slide images may be very large in size.

The proposed compression architecture was compared against a simple CNN-based compression model to measure the improvement in terms of reconstruction.

The results of the comparison reveal that there is significant improvement in the reconstruction using the proposed compression model. The baseline CNN compression model had a PSNR of around 25 dB, while the proposed compression algorithm resulted in a PSNR of about 34.31 dB. As PSNR greater than 30 dB is acceptable in the medical imaging domain, it is clear that there is significant improvement in image reconstruction through the proposed algorithm.

The SSIM values further prove the advantage of using the proposed compression model. With the baseline model, the SSIM value was about 0.89, whereas the SSIM value was about 0.955 for the proposed model. The high SSIM value proves that the images generated through the proposed model maintain higher structural similarity to the source pathology images.

The Mean Squared Error analysis proved this finding. According to the analysis, the proposed compression model generated a much lower reconstruction error, which indicated that it could generate more accurate images when used for image compression.

Loss graphs provide additional proof that the proposed compression model performed better than the baseline CNN model. While the latter reached its maximum performance level quite early in the process of training, the former managed to continue improving its performance and generate lower loss levels in comparison to the baseline model.

This means that the proposed compression architecture could help in maintaining the diagnostic characteristics of the pathology images while compressing their size. It is important to point out that Table 2 provides additional evidence regarding the performance of the proposed compression model.

Table 2. Comparison of compression model performance in terms of PSNR, SSIM, and MSE.

Model	PSNR (dB)	SSIM	MSE
Basic CNN Compression	25.3	0.89	0.003
Proposed ResUNet Compression	34.31	0.9555	0.0004

6.3 Impact of Compression on Network Performance

The last aspect evaluated in the ablation study is the role played by the compression module in determining the performance of the network transmission. Compression would be effective in ensuring that there is reduced traffic flow in terms of the data transmitted over the network, thus enhancing network transmission efficiency.

From the experimentation, it was observed that the compression module leads to the decrease in size of the data by 88.52% for pathology image packages. Consequently, the compression module facilitates the efficient transmission of the pathology images, despite having limited network bandwidth.

From the experiments conducted, it was established that the transmission is successful at 100%, which means that image packages are successfully delivered through the network. The average network throughput observed was about 264.67 KB/second, which is efficient considering the limitations posed by the network.

Table 3. Network transmission performance metrics of the proposed system.

Metric	Value
Transmission Success Rate	100%
Average Throughput	264.67 KB/s
Compression Savings	88.52%
Average End-to-End Latency	1.07 seconds

The mean latency that was found when conducting transmission experiments amounted to roughly 1.07 seconds. Latency is defined as the time spent for compression, transfer, and decompression of an image. Although extra processing effort has been put into the compression algorithm, the system still features relatively low latency of data transmission. The table below shows the network efficiency results.

6.4 Discussion of Component Contributions

The ablation study shows that all components of the suggested telepathology framework play a crucial role in determining the performance of the entire system. With better classification models, the system performs accurately in diagnosis, thus facilitating the ability to classify malignant pathological samples reliably. With the help of better compression models, the system is able to compress images effectively, ensuring the presence of key features needed for analysis. Finally, the delay tolerant model allows the transfer of images in uncertain communication environments.

Combining all these elements into one framework ensures the attainment of a balance between accuracy, quality, and efficient transfer of data. The ablation study reveals that the absence or replacement of any of these components would result in lower system performance.

In conclusion, this study shows that when designing a telepathology system,it is essential to combine artificial intelligence with efficient compression techniques and robust communication protocols. This is evident from Table 4, which gives system performance during multiple transfers.

Table 4. System scalability evaluation under concurrent transmission scenarios.

Concurrent Transfers	Successful Transfers	Avg Latency (s)	Max Latency (s)
5	5/5	6.90	7.17
10	10/10	36.51	36.66
20	20/20	25.51	26.43

7. Discussion

From the outcomes of the experiment, it is evident that the proposed telepathology system can function efficiently based on three key factors: diagnostic accuracy, image compression, and secure data transmission within limited network environments. The combination of the above factors helps to ensure that the system can operate in remote settings when there is a lack of network infrastructures.

The proposed classifier performed efficiently when discriminating benign and malignant histopathology images. The accuracy of the classification model was 99%, while precision and recall rates were nearly 0.99 for both categories. Therefore, it means that the classifier is highly capable of recognizing malignant cases while ensuring that false positives are minimal. The recall value is significant in medical applications since undetected malignant cases might pose challenges to decision-making processes. In addition, the excellent AUC-ROC value of 0.9995 shows that the proposed classifier can efficiently classify histopathological images.

The comparison between the baseline CNN and the proposed approach also indicates the efficiency of the new architecture. While the baseline CNN was able to achieve approximately 90.67% of accuracy, the proposed model was able to achieve approximately 98.98%. From this result, it can be deduced that the new architecture has performed better in learning pathological features from histopathology images.

As indicated in the experiment of image compression, the proposed compression model was able to keep the structure intact while compressing the images of pathologies. The results of 34 dB PSNR value and 0.955 of SSIM value mean that the reconstructed images are highly similar visually to the images before compression. It shows that the compression did not affect their quality, which is required for diagnosis.

In experiments on network transmission, it was shown that the system was able to successfully transmit compressed images of pathologies despite the network constraint. The system was able to achieve a success rate of 100%, throughput of 264.67 KB/s, and end-to-end latency of 1.07 second. It was shown that the delay-tolerant transmission protocol was capable of transmitting the medical images.

In general, the results obtained through experimentation have shown that the application of automated pathology classification, effective image compression, and delay tolerant communication techniques can greatly enhance the feasibility of telepathology applications in bandwidth-limited conditions. With the aid of the suggested architecture, pathologies can be effectively transmitted and analyzed from a distance, thus allowing for more streamlined diagnostic procedures.

8. Limitations and Future Work

Even though the proposed telepathology framework shows excellent performance concerning classification accuracy, image compression quality, and network transmission efficiency, there are certain limitations that should be taken into account when interpreting the findings obtained in this paper.

The first limitation concerns the use of a dataset for evaluating the classification model that includes benign and malignant histopathology images only. Although the model has shown very high accuracy in this binary classification task, practical diagnosis often encompasses other classes as well, such as various types of cancers and tissue states. Multi-class pathology classification would allow diagnosing patients more effectively.

Finally, while the quality of the images compressed by means of the compression model is assessed through the metrics, such as PSNR, SSIM, and MSE, these do not completely reflect the interpretability of images from the diagnostic point of view. Future research may involve clinical evaluations of the image quality based on the assessments made by doctors.

A third limitation is associated with the test environment used to assess the effectiveness of the network transmission algorithm. The experiments involved network constraints to analyze the efficiency of the delay-tolerant protocol; however, a real-world implementation might face other challenges related to network connectivity, bandwidth, and latency issues. It would be interesting to implement the proposed algorithm in an actual clinical setting and measure its performance.

Finally, it should be noted that the system only implements transmission of classified and compressed images; thus, there might be room for improvement concerning transmission efficiency in changing network conditions. In particular, it is possible to implement an adaptive compression strategy to ensure efficient transmissions under changing network conditions.

On the other hand, another possible development direction of the algorithm could be associated with the expansion of its functionality, making it applicable to several institutions that collaborate in improving diagnostic accuracy using federated learning.

Another area for future investigation could be the use of sophisticated image processing technologies that are capable of recognizing distinct pathologies within the histopathology image data, such as tumors, tissues, or cellular abnormalities. This capability can help increase the potential of automated telepathology systems in aiding medical practitioners in their diagnostic decisions.

Even with these shortcomings, the findings from this paper show that the combination of automated classification, intelligent image compression, and delay-tolerant network technology holds promise as an effective way to advance telepathology systems. Further research in this field can lead to better telepathology systems, especially for developing countries.

9. Conclusion

The growing popularity of digital pathology and telemedicine has opened up new avenues for the provision of diagnostic services to more patients in need. Nevertheless, telepathology systems are associated with certain limitations. Specifically, telepathology is often hard to implement in rural or underserved areas due to the poor availability of network infrastructure, as well as issues with efficient transmission of huge-size medical images. To mitigate these limitations, the current study introduced a comprehensive approach for the implementation of telepathology operations. The system integrates automated classification of histopathology images, their compression using deep learning methods, and the use of delay-tolerant communications.

The current research demonstrates that telepathology processes can benefit from incorporating automatic analysis of medical images, as well as their efficient compression. The results show that the presented model exhibits robust capabilities in separating benign from malignant tissue samples, with high values of accuracy, precision, recall, and F1-score. Moreover, the calculated AUC-ROC score of approximately 0.9995 indicates excellent discriminative abilities of the model. The results suggest that automated deep learning algorithms can provide valuable insights into distinguishing abnormal tissue samples.

Additionally, the compression technique implemented within the framework was found to be highly effective in compressing pathology images without losing any visually significant attributes from them. In order to ensure the accurate visualization of various characteristics such as cell structure or tissue organization in the images, high-quality compression becomes extremely vital for histopathological analysis. In terms of performance, it has been noted that the proposed framework produced PSNR values of around 34 dB and SSIM values of approximately 0.955, which proves that the reconstructions remained highly similar to the original images.

Moreover, another valuable contribution offered by the proposed framework relates to the inclusion of a robust delay-tolerant data transmission protocol, allowing for successful image transfer in restricted network settings. According to the results obtained through experimentation, the framework successfully ensured a transmission success rate of 100% while delivering the compressed pathology images to the destination. Additionally, the throughput and latency values obtained suggest efficient use of network resources by the framework.

Furthermore, the scalability tests confirmed that the system will be capable of supporting the simultaneous transmission of multiple images without compromising its performance. Despite several images being transmitted at once, the delay tolerant communication protocol provided for successful delivery of all information packets. The ability to accomplish such a task is especially important in terms of telepathology applications for the distributed environment as several clinics can send their diagnostic images to the pathology center via a single server.

Generally speaking, the results of the presented research prove that the integration of automatic classification, advanced image compression, and reliable communication protocol will ensure the efficiency of telepathology systems in real-life healthcare applications. Moreover, the proposed framework allows for efficient transmission and processing of images while ensuring high accuracy and image quality.

Considering the problem of big-size medical images, insufficient networks as well as issues connected with the automatization of diagnoses, it should be stated that the developed framework is a step towards telepathology platforms' scalability in practice.

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Appendix

Table 5
Summary of Mathematical Notations and Symbols

Symbol	Definition	Symbol	Definition	Symbol	Definition
I	Input histopathology image obtained from the digital pathology scanner	MSE	Mean Squared Error between original and reconstructed images	Tsent	Time at which data transmission begins
Î	Reconstructed image after compression and decompression	MAX	Maximum possible pixel intensity value of the image	Treceived	Time at which data transmission is completed
I_i	Pixel value of the original image at position i	PSNR	Peak Signal-to-Noise Ratio measuring reconstruction quality	Latency	End-to-end delay between sender and receiver
$\hat{I}_i$	Pixel value of the reconstructed image at position i	SSIM(x, y)	Structural Similarity Index between original image x and reconstructed image y	Throughput	Rate of successful data transfer across the network
f(·)	Deep learning classification function	x	Original reference image	Success Rate	Ratio of successful transmissions to total transmission attempts
θ	Model parameters of the neural network	y	Reconstructed image used for comparison	I	Input histopathology image obtained from the digital pathology scanner
y	Predicted class label of the pathology image (malignant or benign)	μx	Mean intensity value of image x		
E(·)	Encoder function in the compression model	μy	Mean intensity value of image y		
D(·)	Decoder function used to reconstruct compressed images	σx	Standard deviation of image x		
z	Latent compressed representation produced by the encoder	σy	Standard deviation of image y		
N	Total number of pixels in the image	σxy	Covariance between images x and y		

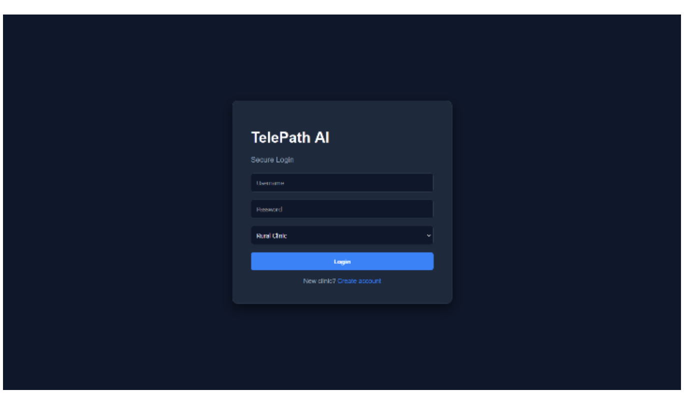

Fig. 10a. We present the secure login interface for accessing the telepathology system.

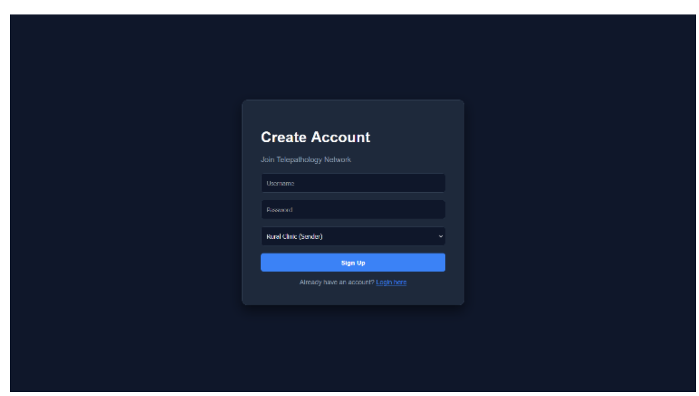

Fig. 10b. We present the user registration interface for onboarding new participants into the telepathology network.